Republic of the Philippines
CAMARINES SUR POLYTECHNIC COLLEGES
Nabua, Camarines Sur

COLLEGE *of* COMPUTER STUDIES

CS 3215 - Human Computer Interaction

LEARNING TASK (DESIGN RULES)

GROUP NO: **9** SECTION: **BSCS 3-A**

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COLICO, Carl Renz M.
MONTENEGRO, John Renzzo C.

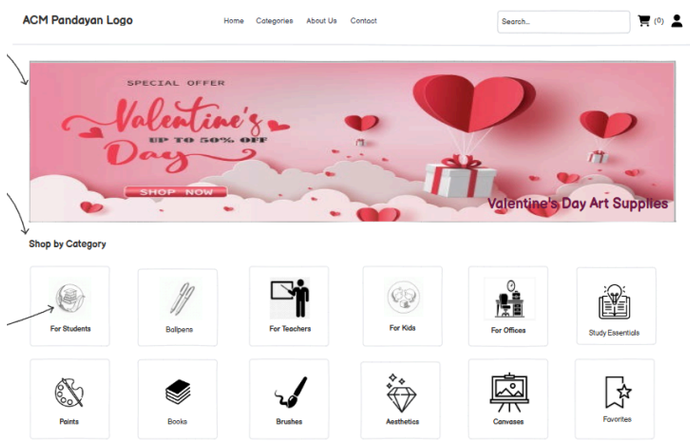
<ACM (AQUINO, COLICO, MONTENEGRO) PANDAYAN BOOKSHOP >

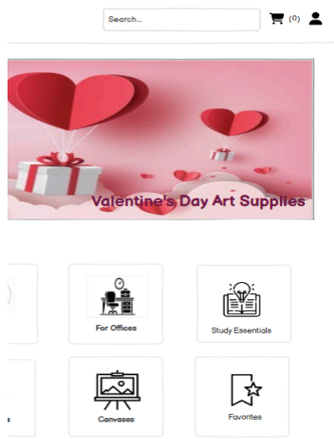
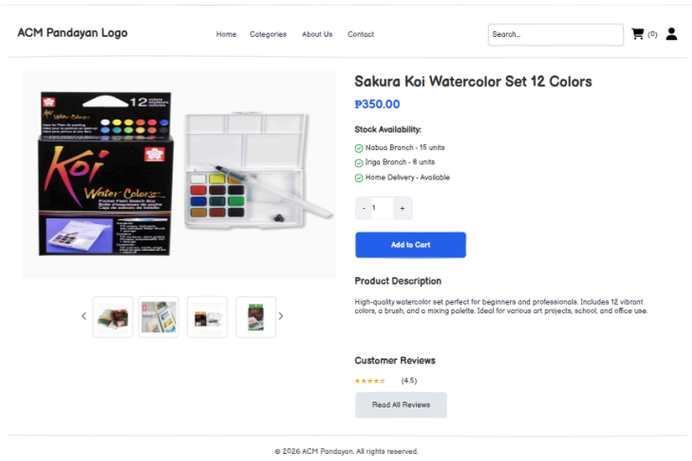
	**PRINCIPLES TO SUPPORT USABILITY**
	**PRINCIPLES OF LEARNABILITY**
	The ACM Pandayan Bookshop redesign incorporates learnability through four key sub - principles. First, Familiarity: the website adopts conventions already like Filipino online shoppers on such familiar icons and navigation tools with Home Categories, Cart and other icons as well as Account icons that mirror popular online shopping platforms. New users immediately recognize where to go because the layout matches their existing mental models of e-commerce. *Second, Predictability: every action course on the e-commerce site yields expected result(s). Clicking add to cart increments the cart badge counter; pressing - proceed to checkout navigates onto another page or tab; submitting an order always produces an order confirmation screen with order ID number. Users can accurately predict the outcome of each interaction without prior training of how it works.* *Third, Consistency: the same navigator bar, set of tabs, color scheme (Pandayan theme), button navigation styles, and set of icons (on every tab(s) or page(s), whether on the Landing page, Search Result(s), or Checkout page flow. Input fields of order always appear in the same format across all forms, reducing relearning effort.* *Fourth, Generalizability: skills learned in one part of the site transfer to others. A user who learns how to search for paper or coloring art supplies can immediately apply the same search - filler pattern to find it without additional guidance.*

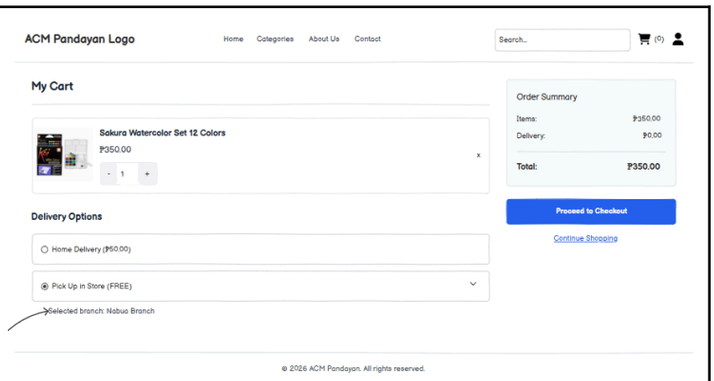
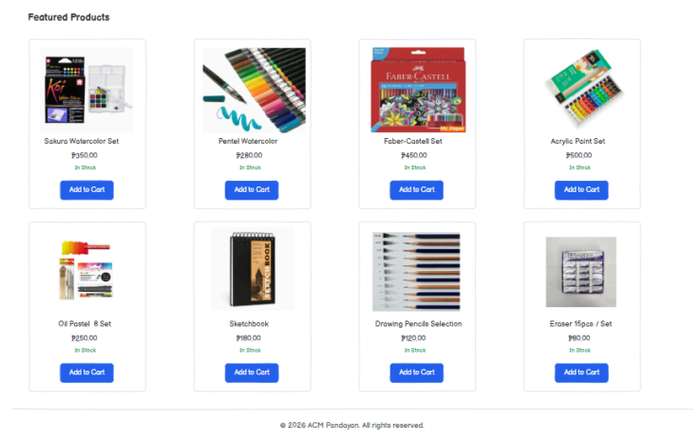
	**PRINCIPLES OF FLEXIBILITY**
	Flexibility is embedded in the ACM Pandayan redesign through multiple pathways and customization options. Dialogue Initiative: the system supports user-driven and system-guided navigation for the users. A user can browse freely by pre-emptive or following the system suggested featured products on the homepage, giving either side control depending on user preference. *Multithreading: users can maintain threads of activity simultaneously. A user can browse items while their cart persists in the background with previously added items, and can open notifications window or panel to check order status updates without losing their current browsing position or current tab.* *Substitutivity: the platform supports equivalent alternatives in both input and output, users can find products via keyword search, category browsing, or the Featured Products section. Payment and Fulfillment can be made via Home delivery or Pick up in store (if address is nearest to physical Pandayan store or if preferred by the user). Each path leads to same goal, giving users the freedom to match interface to their preferred workflow. Customizability: User profile account have personalized dashboard displaying their order history, saved products,accessible profile account and points (if implement a tier points that depends on how user spend) also bulk order features (if institutional or school based). The checkout form pre-fills saved contact details and preferred branch for returning members, reducing repetitive data entry and adapting to the system individual profiles.*
	**PRINCIPLES OF ROBUSTNESS**
	Robustness in the ACM Pandayan redesign ensures users remain informed, in control, and supported throughout every interaction. Observability: the system always communicates its current state to users. The cart icon displays live item count badge, the product details page shows real-time stock availability, and the Order Tracking Page provides a timestamped status timeline (order confirmed, processing, out for delivery, delivered) so users are never left guessing about their order's progress. *Recoverability: the design prevents the irreversible mistakes, On the my cart page, users can adjust quantities or remove items before committing to checkout. On the checkout page, a back navigation arrow allows users to return to the delivery information step if they need to change their selected*

	branch or contact details. Order errors can be reported through the help and support page. *Responsiveness: the mobile optimized design ensures fast page loads and smooth navigation between screens. Confirmation feedback (e.g., 'Item added to cart!' toast notification, order confirmation checkmark screen) is provided immediately after each significant user action so users know their inputs have been registered.* *Task conformance (confirming): the redesigned platform supports all core user tasks end to end: product discovery, add to cart, checkout, payment, order tracking, reordering, bulk orders, point redemption (candidate feature same with loyalty member features) and store location lookup. No task is left incomplete or forced off - the platform.*

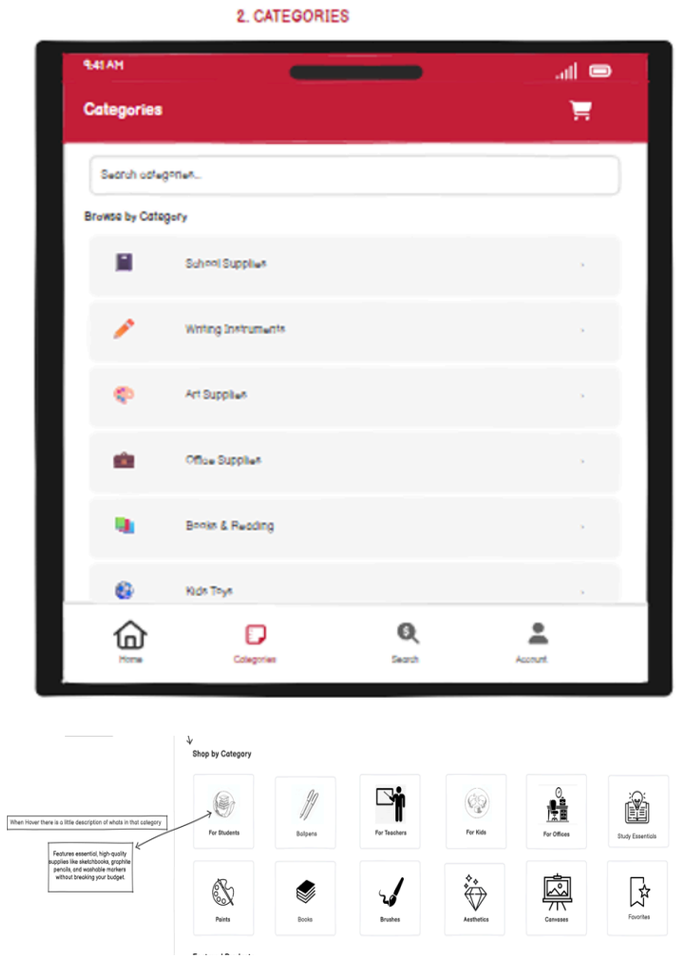
DESIGN RULES APPLICATION ANALYSIS

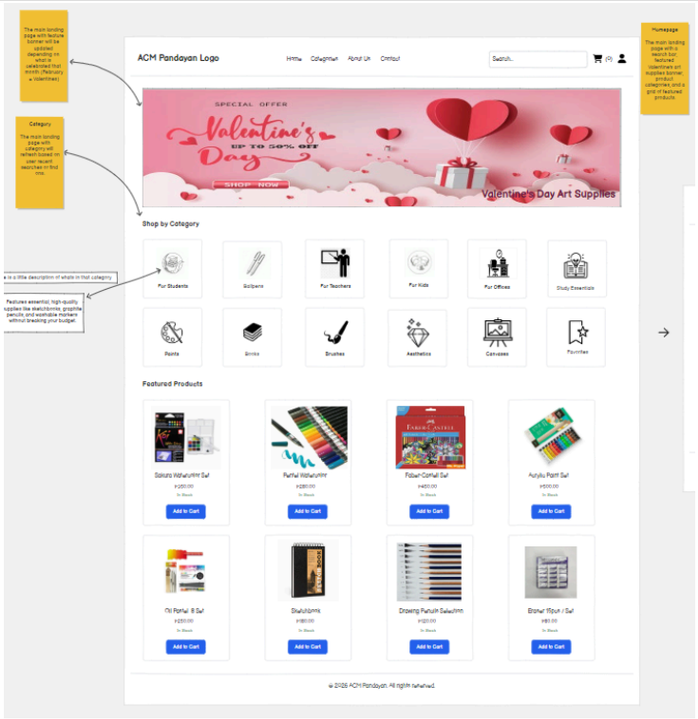
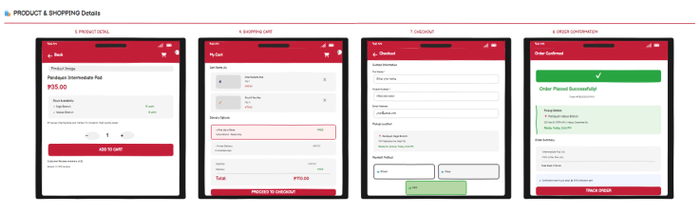
Relevance and application of Interface Design Guidelines and Principles Explanation	Relevance and application of Interface Design Guidelines and Principles Illustration
Context Determines Which Guidelines Apply. Interface design guidelines are not one-size-fits-all rules; their relevance depends heavily on the audience, task, and environment of use. For instance, Nielsen's heuristic 'Match between system and the real world' is especially critical for the ACM Pandayan platform because its primary users (students, teachers, institutional schools) are accustomed to physical bookstore browsing and may have limited e-commerce experience. Applying this guideline means the platform must use Filipino language cues, familiar peso (PHP) pricing symbols, and layout patterns that mirror how items are organized in a physical store.	The Landing Page uses categories labeled 'For Students,' 'For Teachers,' 'For Offices,' and 'Study Essentials' rather than generic labels like 'Category A' or 'SKU Group 1.' These role based models are likely to directly mirror how Pandayan physical stores organized their shelves, making the experience of digital intuitive for first-time online shoppers from the community. 
Visibility of System Status: Users abandon carts when they don't see system updates, which breaks trust in e-commerce. Feedback must come fast and clearly, Mainly for those used to touching products in real stores. Without visible signs of action, like a cart update or payment progress. The shift feels shaky and confusing, and for Pandayan's audience, seeing what happens after clicking is key. Still, many sites miss this basic need, leaving people uncertain. In practice, small confirmations help reduce hesitation more than polished designs ever could. The gap between online ease and physical familiarity remains wide unless the interface speaks plainly to what's happening behind the scenes.	One of the highlights of the design is a shopping cart symbol that never leaves the user's sight, located at the very top right corner of the navbar. Additionally, the counter in the cart updates in real-time (e.g. "(0)"). Besides, the Order Tracking page offers a neat timeline which shows the latest status (e.g. "Out for Delivery"), so the user will always know what's going on with their order.

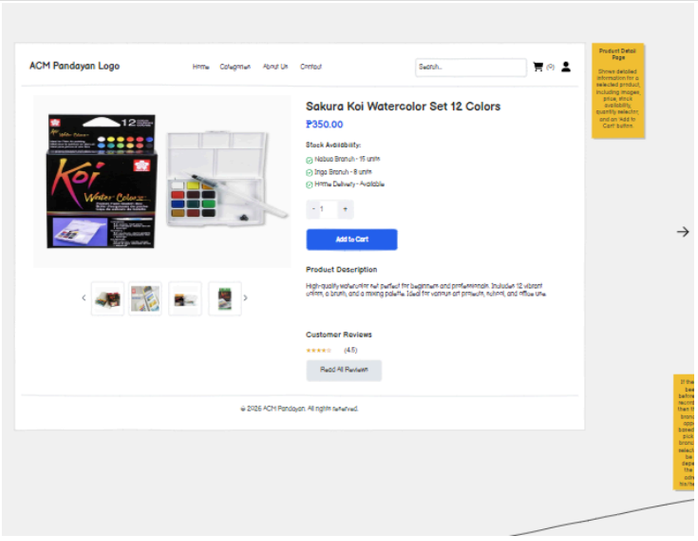
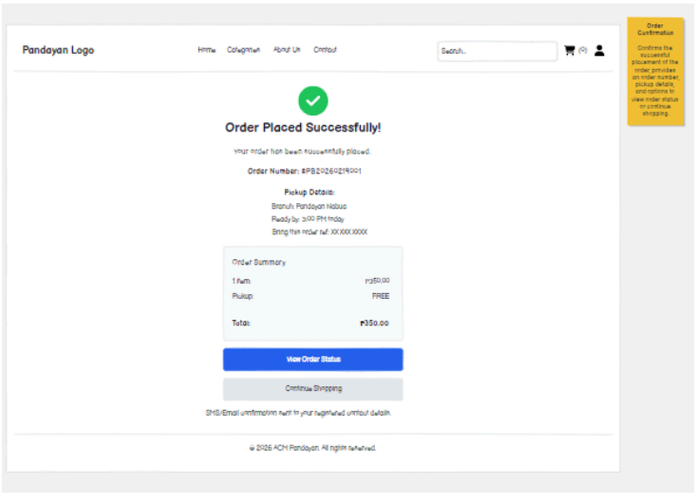
	
User Control and Freedom: Emergency exits are often needed when someone gets stuck in an unexpected part of the system, like a student mid-task or a teacher switching between class orders. These quick paths out prevent users from being lost in layers of confusing menus. So instead of wrestling through long screens, they can simply click back to a safe place. A lot of small decisions happen by accident during busy moments. Turning off actions too late feels risky, so easy reversibility helps ease that tension. People tend to try new things more freely when they know they can undo moves at will.	Items in cart can be quickly changed with "+" and "-" buttons, letting users fix errors before finalizing buy. That said, at checkout, switching between home delivery and store pickup is smooth - no extra steps needed. Still, users feel complete ownership over how orders are fulfilled. As it happens, this setup makes logistics feel personal. Ends up saving time and reducing last-minute stress. 
Error Prevention: Most people mess up when typing fast, Mostly during checkout. Still, fixing errors after they happen feels like playing catch-up. The real solution, while stopping mistakes from forming in the first place. Smart defaults and limited input fields cut down typos before they start. A clean summary shows totals clearly, so users don't have to guess. For Pandayan customers, picking the right branch and double-checking prices stops	On the Delivery Information page, instead of the blank text field, the system is using a dropdown menu for Select Nearest Branch (Pandayan Nabua Branch) . This limited mode of interaction completely eliminates the possibility of users typing a misspelled or non-existent branch. Furthermore, the system displays a clear Order Summary next to the Payment Method so that users are able to check the total costs before they hit the Place Order button.

missed deliveries or wrong charges entirely. In practice, this small effort saves hours of support calls later.	
Aesthetic and Minimalist Design: The primary goal of purchasing school supplies remains clear after the redesign. Instead of overwhelming users with visuals, the site cuts back on extra content that doesn't serve a purpose. As it happens, the original Pandayan site felt like a messy brochure hiding what it was really about. With fewer graphics and more focus, handling now leads directly to product discovery and checkout. That said, each element added has a reason - nothing gets included just for decoration. So the experience ends up feeling calm and easy to follow. The design avoids distractions so users can find what they need fast. With the redesign, the original website's "digital brochure" design has been replaced by a streamlined, grid-based wire-frame structure. The new home page has eliminated visual clutter by including only one prominent, standalone promotional advertisement in addition to several neatly positioned "featured products" that will be the sole focus of the user until he or she uses the purchasing tools.	With the redesign, the original website's "digital brochure" design has been replaced by a streamlined, grid-based wire-frame structure. The new home page has eliminated visual clutter by including only one prominent, standalone promotional advertisement in addition to several neatly positioned "featured products" that will be the sole focus of the user until he or she uses the purchasing tools. 

Cognitive Psychology Facilitation Explanation	**Cognitive Psychology Facilitation Illustration**

Heuristics Support Recognition Over Recall The Recognition Rather Than Recall: Cognitive psychology establishes that recognition is significantly easier than recall because it involves matching a stimulus to memory rather than retrieving it from scratch. Heuristics codify this into design guidance: use visible options, labels, and icons rather than requiring users to remember commands, codes, or navigation paths. This is especially relevant for the Pandayan platform's target users, many of whom are infrequent online shoppers.	The Categories Page uses icon-labeled tiles (e.g., a pencil icon for 'Writing Instruments,' a palette icon for 'Art Supplies,' a briefcase icon for 'Office Supplies') rather than a plain text dropdown. Users recognize the icons through visual matching rather than having to recall category names, dramatically lowering the cognitive effort required to begin product discovery. ILLUSTRATION 1  ILLUSTRATION 2
Mental Models and Conceptual Mapping: Cognitive psychology underlines that users create mental models on the basis of previous experiences. By fitting an interface with these expectations, users are able to explain and navigate the system more effectively without getting lost. In the case of the Pandayan platform, numerous users have already experienced physical book stores and typical e-commerce designs. The system is able to replicate these structures	The layout of the Home Page is designed after the physical store by organizing the products into categories like; For Students, For teacher and office supplies. This is similar to the way things are laid out in real Pandayan stores where people can learn to use the store by intuitively browsing the categories without having to learn a new system.

digitally, and with such replication, cognitive friction is minimized, and users can effortlessly apply real-world knowledge to the online context.	
Cognitive Load Reduction: The level of mental effort to utilize a system is called cognitive load. Interfaces, which reduce the unwarranted complexity, enable users to concentrate on their core activities. The Pandayan redesign eases the cognitive load by making navigation easier, minimizing clutter, and showing only pertinent information at each step of the shopping experience. This is particularly critical to the users that might not be intensive online shoppers.	The Payment System is broken down into easy to understand step by step pages (Cart → Delivery Information → Payment → Confirmation). The best thing is that only required inputs are shown on each screen, thus avoiding the potential overload of users and allowing them to accomplish tasks effectively. 
Attention and Visual Hierarchy: Human attention is selective and initially the user will concentrate on the elements that are visually prominent. Good interfaces direct the attention of the user by appropriate application of size, contrast, spacing and positioning. Pandayan platform uses visual hierarchy to emphasize significant actions and information so that the user will not have to spend much time on understanding what to do next.	The most important actions like Add to Cart and Proceed to Checkout are presented in big high contrast buttons, whereas other secondary information like product details are placed below or at the side of them. This makes sure that users instinctively concentrate on important activities required to make a purchase.

	
Feedback and Reinforcement: Cognitive psychology emphasizes the significance of instant feedback in the process of strengthening the user behavior and lessening the uncertainty. By getting positive feedback that whatever they are doing works, users gain confidence and will tend to keep engaging with the system.	Once a user presses the button to add something to the cart, a confirmation message, and a new cart badge (example : 0), should be seen. Likewise, the Order Confirmation page shows explicit information (Order Number, Payment Status, Pickup Time) to support the fact that the transaction is successful. 

TAKEAWAYS

Name of Member: **AQUINO**, Sean Xander B.
Contribution to the Group: **Relevance and application of Interface Design Guidelines and Principles (Rows 2, 3, 4, and 5)**
Learnings: Upon completing a learning exercise involving designing and developing a user interface, I had the following successes: I learned that a good user interface should be flexible and adaptable to the different types of shoppers who are likely to be using it. I also learned that flexibility and robustness along with providing multiple pathways to complete a goal (searching vs. visually selecting categories) is imperative for the user. In addition, I learned from my redesign of the checkout process at Pandayan that it is important to proactively help correct user errors from the onset (e.g., utilize dropdowns for store location instead of open text fields). In the end, building trust in a robust system involves giving the user information and control, along with limiting the possibility of creating an irreversible error.

Name of Member: **COLICO**, Carl Renz M.
Contribution to the Group: **Principles of Learnability**, **Flexibility, Robustness**, **Relevance and application of interface design (Rows 1)**, **Cognitive Psych. (Rows 1)**
Learnings: This Learning task given to us deepened my understanding of how learnability shape early work or planning design of website or any decision on the making. I realized that making an interface learnable is not just about adding familiarity into the layout itself but through consistent navigation, icons, buttons and other generalizable patterns. Connecting these principles to the AMC Pandayan made it feel practical and relevant.

Name of Member: **MONTENEGRO**, John Renzzo C.
Contribution to the Group: **Cognitive Psychology Facilitation (Rows 2, 3 4, 5)**
Learnings: In this activity I learned that learnability provides ease of use to new users. The redesign renders the site user-friendly through the familiar layout, predictable actions, consistent design components, and transferable skills across the sections. The system is easy to use as users can learn and navigate without any training.

Mimicking the usual e-commerce interfaces, the users understand right away how to interact with the site, and the time to learn is minimized.